

Contact

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Top Skills

Artificial Intelligence (AI)

Terraform

AWS Lambda

Certifications

Data Science and Analytics using R programming

AWS Lambda Foundations

Amazon Web Services Cloud Practitioner

Prediction and Control with Function Approximation

Terraform for AWS - Beginner to Expert

Shayan Shafquat

GenAI @ Allstate | RL | Cognitive Neuroscience

London, England, United Kingdom

Summary

I am a Senior Data Scientist with 2.5+ years of experience specializing in fine-grained pattern recognition, engineering LLM security solutions, and deploying AI (GenAI & Classical ML) approaches across insurance, transportation, and e-commerce industries. My foundation in Applied Mathematics and Computational Neuroscience has enabled me to contribute to projects at the intersection of AI and decision neuroscience.

Currently, I focus on developing LLM-based applications, leveraging multi-agent systems and RAG frameworks to optimize workflows and improve AI security. At Allstate, I design and manage Lambda-triggered infrastructure for real-time data processing and enhance LLM-powered systems with advanced evaluation frameworks and contextual grounding using Guardrails.

My technical expertise spans machine learning (multi-pass LLM frameworks, RAG systems, deep learning in NLP, CV, RL), responsible AI practices, and cloud infrastructure, including AWS services (Lambda, S3, Bedrock, SageMaker) and serverless architecture. I also specialize in backend development, including FastAPI/Flask API development, system architecture design, and Docker containerization.

With proven business impact, including a 5% increase in cashless rides and improvements in conversion prediction and loss capture rates, I am committed to pushing the boundaries of AI-powered applications and building robust security measures to defend against prompt injection and adversarial attacks.

Notably, I have worked on Neural Dynamics and Disease Modeling, where I simulated Alzheimer's disease progression and explored neural stability during NREM sleep. I've also contributed to converting large-scale cortical models, ensuring compatibility and accuracy in simulating brain networks.

Experience

Allstate

Senior Data Scientist

January 2025 - Present (7 months)

Northern Ireland, United Kingdom

Enkrypt AI

AI Research Consultant

June 2023 - September 2023 (4 months)

- Developed a malicious file scanner for python scripts and jupyter notebooks to enhance ML lifecycle security.
- Built a dataset for prompt injection and backdoor attacks to fine-tune LLMs for adversarial robustness.
- Implemented demo-ready LLM security using RAGAS and NeMo GuardRails to counter toxicity, bias, and jailbreaks.

Google Summer of Code

Open source, cross simulator, large scale cortical models in NeuroML

June 2022 - September 2022 (4 months)

- Translated Layer 5 pyramidal cell properties to NeuroML, ensuring model accuracy.
- Implemented GitHub Actions for maintaining model integrity in development.
- Collaborated with experts to adhere to computational neuroscience standards for reproducible simulations.

Ola

Data Scientist I

September 2020 - January 2022 (1 year 5 months)

Bengaluru, Karnataka, India

Led the development of a user identification system at Ola, increasing cashless ride transactions by 5%. Improved peak pricing accuracy by 2.2% and automated user-level pricing during COVID-19. Collaborated to enhance in-house data preprocessing utilities for AWS S3 retrieval.

Ola (ANI Technologies Pvt. Ltd)

Internship (Research Engineer)

May 2019 - July 2019 (3 months)

Bangalore

- Engineered features from partner ride data to predict login hours, optimizing incentive strategies.
- Conducted A/B testing to validate incentive optimization effectiveness across multiple cities.

Prof. Sameer Mathur, IIM Lucknow

Intern (Data Analytics with Managerial Applications)

May 2017 - June 2017 (2 months)

- Completed the internship under the guidance of Prof. Sameer Mathur (Professor, Indian Institute of Management (IIM), Lucknow)
- Learnt Regression Modelling, Ordinary Least Squares(OLS), various tests for hypothesis verification and Inferential Statistics.
- Got introduced and mastered the concepts of R programming language

Education

University of Nottingham

Master of Science - MS, Computational Neuroscience, Cognition and AI · (September 2023 - December 2024)

Indian Institute of Technology, Kharagpur

Integrated Msc, Mathematics and Computing · (2015 - 2020)